

NEPAL ENGINEERING COLLEGE
FINAL ASSESSMENT

Level: Bachelor
Programme: BE
Course: Cloud Computing and Virtualization
Semester: VII

Year : 2026
Full Marks: 100
Pass Marks: 45
Time: 3Hrs

Students are advised to provide clear, concise, and relevant answers. Responses must directly address the question asked. Unnecessary elaboration or deviation from the topic may not be considered for evaluation of the answer.

The figures in the margin indicate full marks.

Attempt all the questions.

1. (a) Explain the NIST cloud computing reference architecture with block diagram and give real-world example of its actors. 8
(b) Explain each cloud deployment model, highlighting the advantages and limitations of each. 7
2. (a) Define a hypervisor. How it is different from operating system? Compare Type 1 and Type 2 hypervisors based on architecture, performance, and use cases. 8
(b) Explain the concept of container and containerization. Compare containers with virtual machine. 7
3. (a) Define Virtual Private Network. How does a VPN provide secure communication over the internet? Explain the roles of encryption and tunneling. 7
(b) What is a DNS service? Explain its main purpose in translating domain names to IP addresses. Describe how a DNS service works when a user types a website URL in a browser. Include the role of recursive and authoritative DNS servers. 8
4. a) Explain how cloud storage services manage data based on data type and frequency of access. List out the key characteristics of each cloud storage service. 8
b) Define Database as a Service (DBaaS). Explain the major services provided by DBaaS and analyze the challenges associated with its implementation. 7

5. a) What is a message digest? Explain how the hashing mechanism works in cloud security. Discuss how hashing and message digests ensure message authenticity and data integrity. 8

OR

Compare Layer 4 and Layer 7 load balancers in terms of their working and decision-making criteria. For each of the following use cases, identify whether a Layer 4 or Layer 7 load balancer is more appropriate and justify your choice with valid reasons:

- i. A high-traffic online gaming service that requires fast packet forwarding with minimal latency.
- ii. A cloud-based web application that needs request routing based on URLs, HTTP headers, or user sessions.

- b) Define a Service Level Agreement (SLA) in cloud computing. Explain different types of SLAs commonly used and outline the lifecycle of an SLA. 7

6. a) Define Identity and Access Management (IAM). Explain the role of IAM in ensuring cloud security. How can weaknesses in IAM lead to security threats, and how can these threats be mitigated? 8

- b) What is Public Key Infrastructure (PKI)? Explain its components and how PKI is used to manage digital certificates and public/private keys in cloud computing. 7

7. Attempt any two.

- a) Jericho Cloud Cube Model
- b) Single Sign-On
- c) Distributed File System

5*2
=10

NEPAL ENGINEERING COLLEGE

Level: Bachelor
Programme: BE

Semester – Spring

Year : 2026

Full Marks : 100

Pass Mark : 45

Time : 3 hrs

Course: Engineering Economics (Computer 7th)

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1.
 - a. Define Engineering Economics. Why engineering economics is considered as important aspect for making decision for engineers? Explain. 8
 - b. What is nominal and effective interest rate? If you have Rs. 1000000 loan now from a bank, how much Rs. should you pay as instalment per two month for 5 years if bank interest rate is 12 % per year? 7
2.
 - a. Define time value of money. Dinesh an electronics and communication engineer is planning to place a total of 20% of his salary, which is Rs. 250000 per year. He expects 7% increase in salary for next 15 years if the mutual fund results in 10% annual return, what will be the amount at the end of 15 years. If salary increases by Rs. 25000 per year, what will be the amount? 7
 - b. Define sunk cost and opportunity cost. Explain direct cost and indirect cost with suitable example. 8
3.
 - a. What are the drawbacks of IRR methods? Evaluate the IRR for the following project and decide whether the project is accepted or not? Also draw the investment balance diagram. 7

Initial Investment (Rs.)	100000
Annual Revenues (Rs.)	62100
Annual Expenses (Rs.)	30000
Salvage Value	30000
Project Life (Years)	5
MARR	10%

- b. Consider the following three sets of mutually exclusive alternatives: 8

Alternatives			
EOY	Project X (Rs.)	Project Y (Rs.)	Project Z (Rs.)
0	-2000	-1000	-3000
1	1500	800	1500
2	1000	500	2000
3	800	500	1000

Which project would you select based on BCR method on incremental investment assuming that MARR=15%.

OR

Find the B/C ratio by both conventional and modified methods using AW

formulation.

First investment = Rs. 1,00,000
 Annual revenue = Rs. 40,000
 Annual cost = Rs. 20,000
 Salvage value = Rs. 25,000
 MARR = Rs. 11%, Number of years = 8

4. a. Make a selection from the following two mutually exclusive alternatives:

Alternatives	A	B
Investment (Rs)	450000	600000
Annual revenue (Rs)	22000	26000
Annual expenses (Rs)	7450	11020
Salvage value (Rs)	25000	28000
Useful life (years)	6	9

Using repeatability assumption method, MARR=10% per year.

OR

A multipurpose hydroelectric project under consideration of the government, whose estimated benefits and costs expected to be derived from the project, are listed as below:

End of year	Annual cash flow (Rs.)
Initial Cost	180000000
Annual power sales	12000000
Annual flood control saving	5000000
Annual irrigation benefits	8000000
Annual recreation benefits	4000000
Annual operating and maintenance costs	5000000

Suggest, based on **Annual worth method**, the government about implementing the project of life 45 years. MARR = 10%.

- b. Calculate the future worth of the following cash flow deposited at 10% compounded monthly for 5 years.

- Rs. 60,000 at the end of each year
- Rs. 60,000 at the beginning of each year

OR

Perform sensitivity analysis of the following project over range of $\pm 30\%$ at the interval of $\pm 10\%$ in:

Initial investment, Net annual revenue, Salvage value, interest rate and Useful life; use present worth formulation. Draw also sensitivity graph.

Capital Investment (Rs.)	11500
Annual Revenues (Rs.)	5000
Annual Expenses (Rs.)	2000
Salvage Value	1000
Project Life (Years)	6
MARR	10%

5. a/ What do you mean by depreciation, corporate tax and book value? Consider

the following accounting information for a computer system:

Cost of the asset = Rs. 10000

Useful life = 5 years

Salvage value = Rs. 100

Determine the optimal time to switch from double declining balance to straight line depreciation and the resulting depreciation schedule.

OR

Evaluate the ERR for the following project and decide whether the project is accepted or not?

Initial Investment (Rs.)	100000
Annual Revenues (Rs.)	62100
Annual Expenses (Rs.)	30000
Salvage Value	30000
Project Life (Years)	5
MARR	10%

- b. Calculate both types of payback period for the given information:

Initial investment = Rs. 6000

Annual benefit = Rs.3000

Annual O&M cost = Rs. 1000

Salvage value = Rs. 1500

MARR = 10%

Useful life = 5 years

Also check the feasibility of the project (Take standard payback period = 3 years).

6. a. Explain different project funding mechanism in Nepal for public projects.

OR

✓ Explain different types of cash flow with examples.

- b. What are the methodologies of Engineering Economics for Project Selection

7. Write short notes on (Any Two): 2×5

- Independent and contingent project
- Effective interest rate vs nominal interest rate
- ✓ Supply and Demand

NEPAL ENGINEERING COLLEGE

Level: Bachelor

Programme: BE

Course: Data Science and Analytics.

Year : 2026

Full Marks: 100

Pass Marks: 45

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. (a) Explain the Knowledge Discovery process from data (KDD) with its major phases. How does data integration address the challenge of combining the data from multiple sources, multiple formats and forms? 8
- (b) Differentiate between Structured and Unstructured Data and Distinguish Between Numeric and Categorical Data. 7
2. (a) Explain Pearson correlation and how it measures the relationship between two numeric variables? Show it by considering an example. 8
- (b) Describe the Empirical Distribution of Numeric Data. Explain Histogram, Normal Distribution, Exponential Distribution and Power Law Distribution with Examples. 7
3. (a) Explain Multivariate Linear Regression with its Matrix Formulation. Explain R squared and R squared adjusted with their formula. 8
- (b) You are given the following two-dimensional datasets consisting of 6 observations: (2,1), (3,5), (4,3), (5,6), (6,7), (7,8). Using the Principal Component Analysis (PCA) algorithm, compute the first principal component of the dataset. 7
4. (a) You are given the following data. 8

x	1	2	3	4
y	-2.3	0.5	9.3	20.8

Assuming that there is an unknown functional relationship $y = f(x)$, estimate the value of y at $x = 2.9$ using the Nadaraya-Watson regression estimator. Take bandwidth $h = 0.5$.

- (b) What is the importance of bandwidth in kernel regression? 7
How is it selected? Explain with Example.
5. (a) Describe CART algorithm for decision tree with example. 8
Explain how Logistic Regression is applied in classification Problem.
- (b) Apply the K-means Algorithm with $K=2$ on the following Dataset: (160,55), (175,65), (172,60), (180,70), (178,68), (165,58). 7
6. (a) What is time series analysis? Explain its components in detail. 8

OR

- What conditions must be satisfied to conclude that event A causes event B? Describe why granger causality is not considered a true causality.
- (b) Define ARMA and ARIMA models. What is the role of AIC and BIC in model selection? 7
7. Write Short Notes on (Any two):
- (a) Permutation tests for Variable Importance
 - (b) Zipf's Law
 - (c) Multi-collinearity and variance-inflation factors (VIF).
- [10]

NEPAL ENGINEERING COLLEGE

Level: Bachelor

Semester: Fall

Year : 2025

Programme: BE

Full Marks: 100

Course: Entrepreneurship and Professional Practice

Pass Marks: 45

(7th sem computer)

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define entrepreneurship. Explain the terminologies: startup, enterprise, entrepreneur, entrepreneurial attitude, entrepreneurial intention, and entrepreneurial behavior. Also, compare entrepreneurship with employment using suitable points. 8
b) Discuss the emerging technologies influencing entrepreneurship. Explain the global and local startup trends, with reference to the present business environment. 7
2. a) Explain the sources of new business ideas. Describe the tools and techniques used for idea generation with suitable examples. 8
b) Explain environmental analysis in entrepreneurship. Discuss the components of an entrepreneurship ecosystem, highlighting the supports and barriers for startups. 7
3. a) Discuss the importance of team building and SMART goals for the success of a startup. Suggest strategies to recruit co-founders and early employees. 7
b) Develop a Go-To-Market (GTM) strategy for a hypothetical product, including target market, channels, and promotional activities. 8
4. a) Compare and contrast business model canvas, value proposition canvas, and lean canvas. Explain in which scenarios each canvas is most effective. 7
b) Critically evaluate the role of a business pitch in securing funding. Explain why pitch delivery skills are as important as the content. 8
5. Read the following case study and answer the questions below: 15
ABC Pvt. Ltd., a software company in Nepal, is developing a school management system. The senior project manager, Mr. Ram, is

responsible for approving the software tools and cloud services for the project. During the procurement process, he favors a particular cloud service provider, which happens to be partially owned by his brother. Several engineers in the team, including Shyam, notice that other providers are more cost-effective and secure, but Mr. Ram insists on using his brother's company.

Shyam is concerned that using a less suitable provider could compromise the project's security and reliability. He evaluates the options carefully and documents the risks associated with Mr. Ram's preferred provider. After gathering evidence, he reports the issue to the ethics committee to ensure the best decision is made for the project.

The project team ultimately switches to a more secure and efficient provider, ensuring that the system works safely for students and staff. The company's initiative to provide free IT workshops in rural schools remains supported.

Questions:

- I. How did Shyam demonstrate professional integrity while handling the situation?
 - II. What challenge did Shyam face while recommending the cloud service provider, and why was it important to handle carefully?
 - III. Suppose the ethics committee ignored Shyam's report. What could be the short-term and long-term consequences for the project?
 - IV. How does making the right decision in this situation reflect on the company's responsibility toward the community?
 - V. How does documenting risks and decisions help maintain transparency and accountability in projects?
6. a) A software company develops a new AI-based educational app. Identify which parts of the software can be protected under copyright, patent, design, trademark, and trade secrets. Explain why each type of protection applies to the respective part. 7
- b) Analyze the risks a startup faces if it ignores business registration or taxation laws in Nepal. How can proper legal compliance protect the startup's reputation and growth? 8
7. Write short notes on: (Any two) 2×5
- a) Value Proposition and Minimum Viable Product (MVP)
 - b) TAM, SAM, SOM
 - c) Business plan
